Fuel Efficiency Prediction Using Regression-Based Machine Learning: A Case Study on Vehicle Fuel Economy Data

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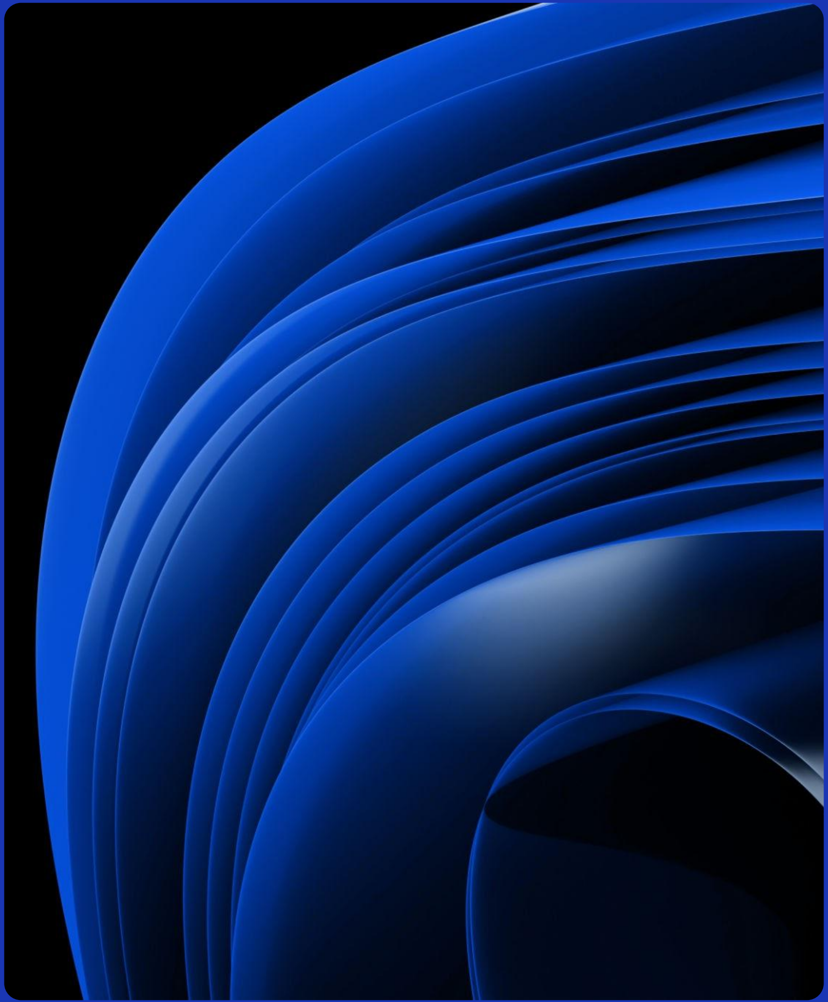
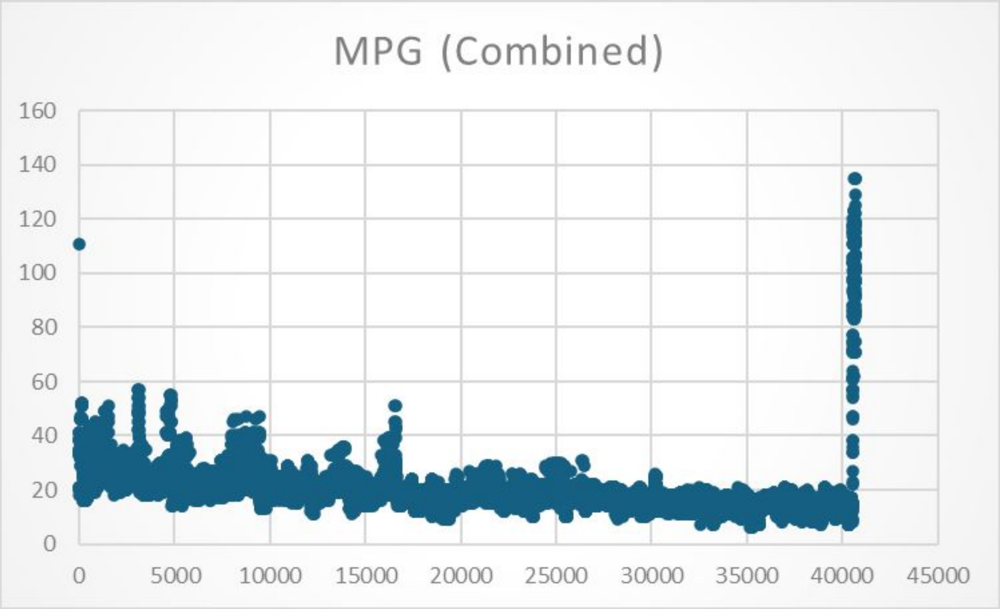


Dataset

Vehicle fuel Economy, data.gov

contains fuel economy data that was obtained from vehicle testing as well as by vehicle manufacturer.

- 40,705 vehicles
- 93 Attributes
- 26 brands
- 1984 - 2019

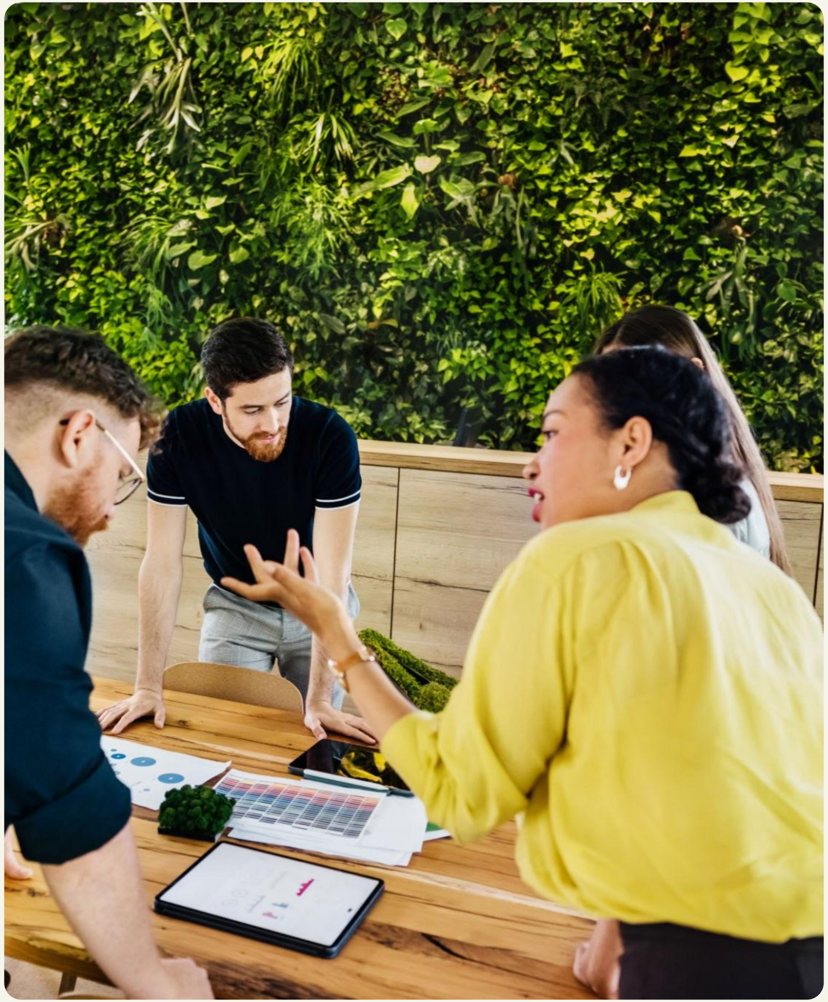


Article

Article: Linear regression for predicting mpg

Author: Khalid Jeffal

This article from 2021 uses linear regression model from Machine learning to predict a vehicle's fuel efficiency (MPG), it also includes guides on data visualization.



Proposed Model

Our goal was to create our own regression model to predict MPG based on vehicle attributes such as engine horsepower, displacement, and number of cylinders while using the article as a reference to guide us along the way.

Then we could compare our prediction with the actual dataset using metrics like RMSE or MAE (mathematical metric to show how close our prediction was). The article also uses RMSE so we had a ballpark estimate of what to aim for.

Proposed Model

Data preprocessing

Eliminate missing, noisy and inconsistent data by removing non relevant attributes, null values, and outliers.

Calculate correlation values.

Algorithm

Implement a multiple linear regression model to evaluate the relationship between independent variables and the target variable.

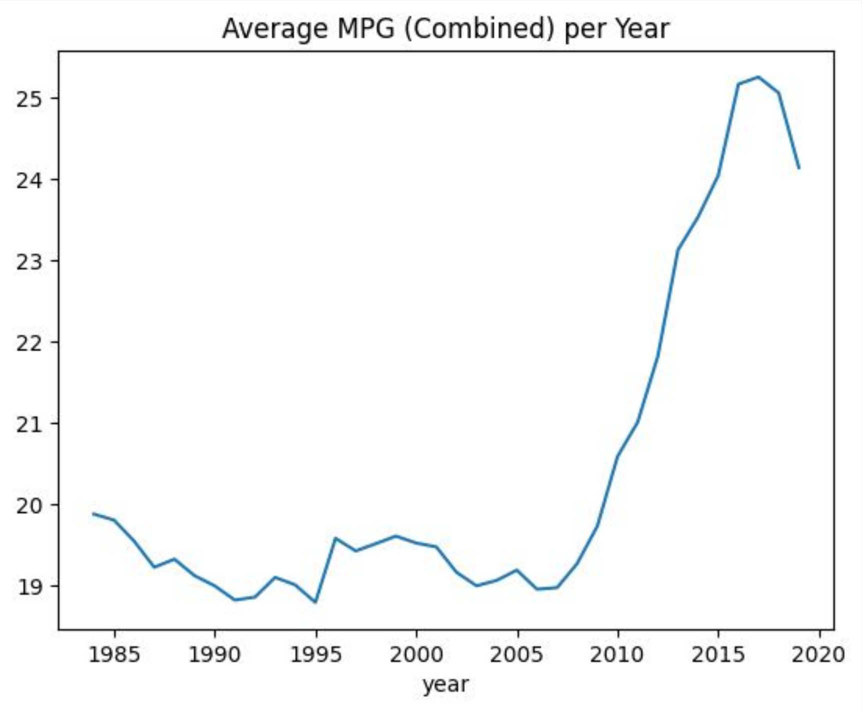
Train model

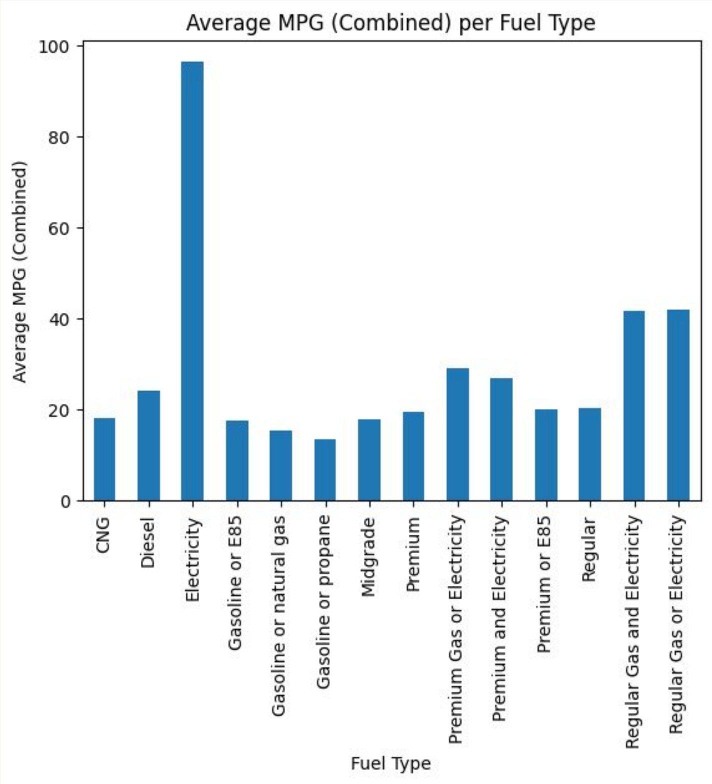
Train the model with a subset of data points (70 %)

Evaluate

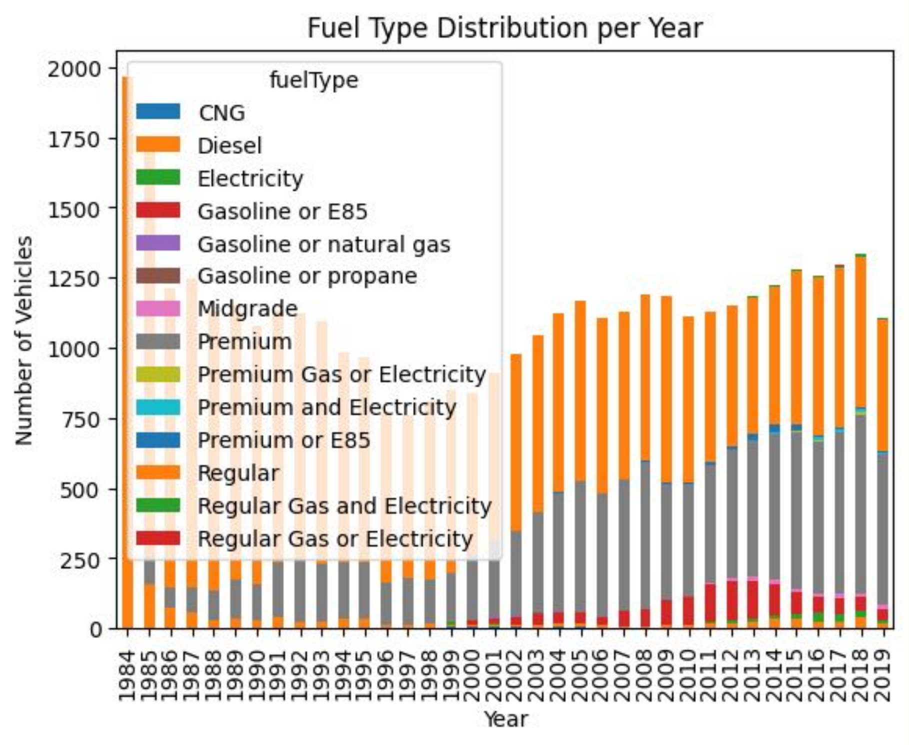
Compare predictions with the remaining data points and calculate evaluation metrics.

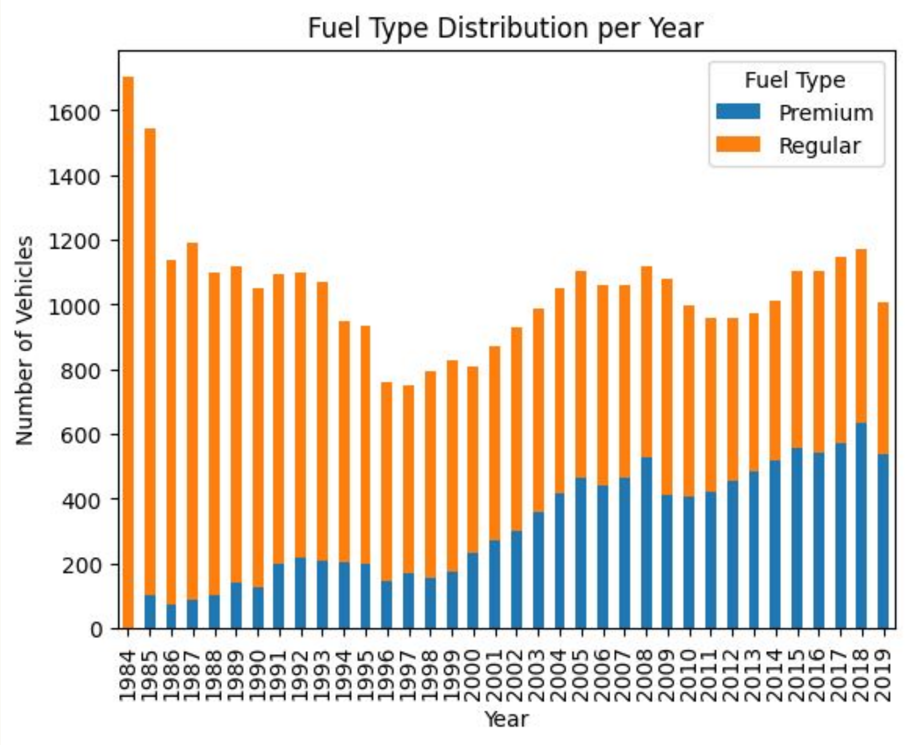
Exploring the Dataset



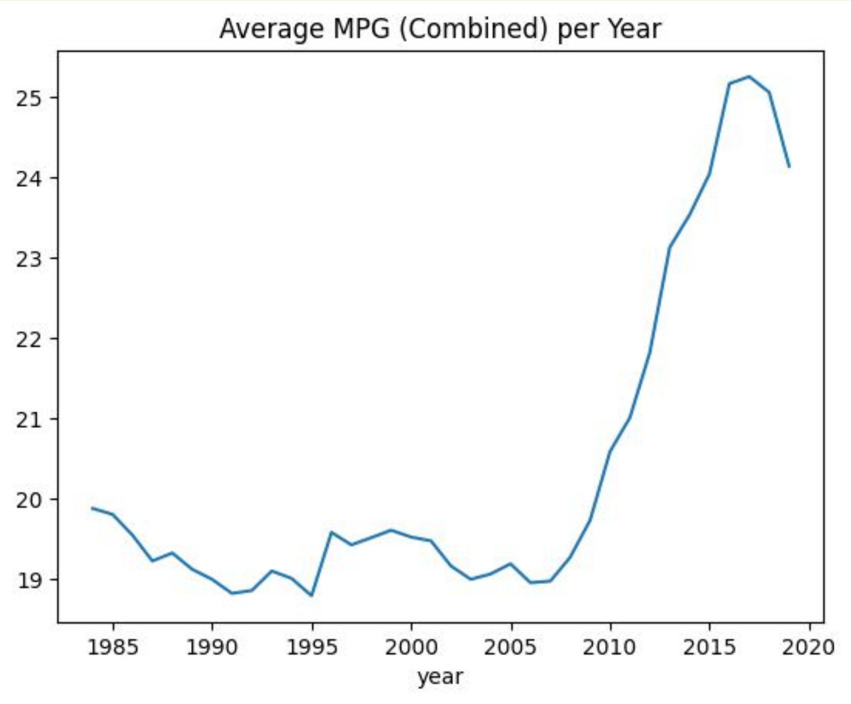


Exploring the Dataset

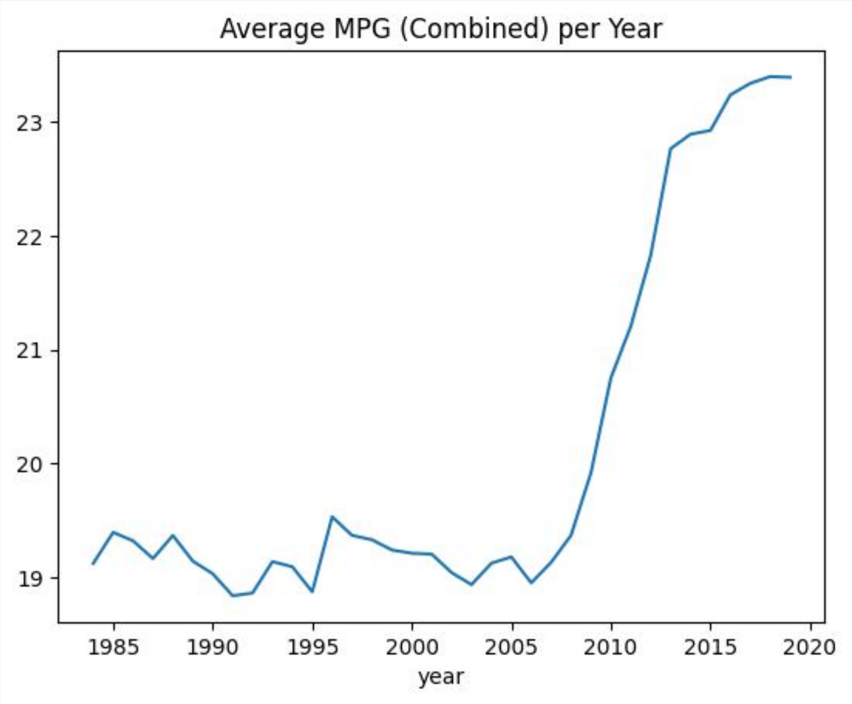




Original (All Fuel Types)



Reduced (Regular, Premium)



THANK YOU!